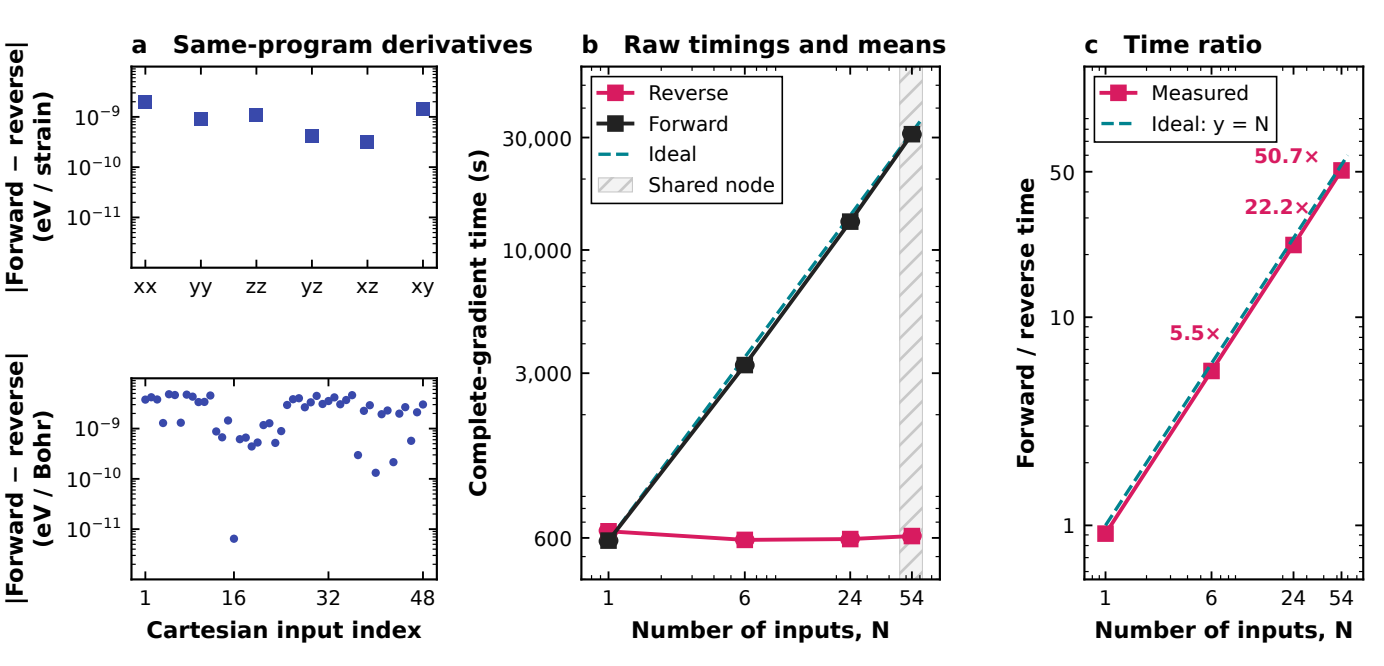
a Same-program derivatives
|Forward − reverse| (eV / strain)
10^{-9}
10^{-10}
10^{-11}
xx
yy
zz
yz
xz
xy
|Forward − reverse| (eV / Bohr)
10^{-9}
10^{-10}
10^{-11}
1
16
32
48
Cartesian input index
b Raw timings and means
Reverse
Forward
Ideal
Shared node
Complete-gradient time (s)
30,000
10,000
3,000
600
1
6
24
54
Number of inputs, N
c Time ratio
Measured
Ideal: y = N
50.7×
22.2×
5.5×
Forward / reverse time
50
10
1
1
6
24
54
Number of inputs, N